Consensus Multi-Layer Edge Combination for Scientific Paper Clustering

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Abstract

We propose a multi-layer edge combination pipeline for CPM Leiden clustering of scientific paper networks. The pipeline combines citation-based edges (direct citation, bibliographic coupling, co-citation) with embedding-based edges using a *consensus* strategy: edges confirmed by multiple independent layers receive multiplicative weight amplification. Combined with per-node top-k filtering, 1/rank normalization, and automatic resolution (γ) selection, the pipeline produces balanced cluster distributions across fields with varying citation densities. Experiments on two OpenAlex fields (Chemical Engineering, 115K papers; Arts & Humanities, 754K papers) demonstrate that (1) consensus combination assigns boundary nodes more accurately than simple summation (29:26 in 55 LLM blind reviews), (2) adding embedding edges reduces cluster count by 60% while doubling average cluster size, and (3) automatic γ search is essential as optimal resolution shifts 24–32$\times$ between 3-layer and 4-layer configurations.

1 Introduction

Clustering scientific papers into research communities requires constructing a similarity network from multiple heterogeneous signals. Citation-based measures—direct citation (DC), bibliographic coupling (BC), and co-citation (CC)—capture different aspects of scholarly relatedness but have uneven coverage: BC may cover only 49% of papers in humanities fields, while DC is universally available but sparse. Embedding-based similarity (e.g., SPECTER2 k-NN) provides universal coverage but may introduce noise from stylistic rather than topical similarity.

The central question is: *how should these heterogeneous layers be combined into a single network for community detection?*

We address three sub-problems:

1. **Normalization**: Edge weights across layers have incomparable scales.
2. **Combination**: How to merge normalized layers into one edge table.
3. **Resolution**: The CPM resolution parameter γ must adapt to the combined network's density.

2 Method

2.1 Pipeline Overview

2.2 Top-k Filtering (Step 1)

Each layer independently retains only the k strongest neighbors per node (symmetric mode: edge kept if *either* endpoint has it in their top-k). This removes long-tail noise that otherwise dominates after normalization. With k=30, cross-cluster edges drop from 70% to 31% of the combined network.

Algorithm 1 Consensus Multi-Layer Combination

Require: Edge tables $\{E_\ell\}_{\ell=1}^{L}$ for L layers, target $\max_{\%}$

for each layer ℓ **do**

  $E'_\ell \leftarrow \textsc{TopK}(E_\ell, k{=}30)$ {per-node top-k filter}

  $E''_\ell \leftarrow \textsc{RankNorm}(E'_\ell)$ $\{w_{ij} \leftarrow 1/\mathrm{rank}(w_{ij})\}$

end for

$E_{\text{combined}} \leftarrow \textsc{ConsensusSum}(\{E''_\ell\})$ {Eq. 1}

$E_{\text{gcc}} \leftarrow \mathrm{GCC}(E_{\text{combined}})$ {giant connected component}

$\gamma^* \leftarrow \textsc{AutoGamma}(E_{\text{gcc}}, \max_{\%})$ {binary search}

$\mathcal{C} \leftarrow \textsc{Leiden}(E_{\text{gcc}}, \gamma^*)$

$\mathcal{C}' \leftarrow \textsc{Postprocess}(\mathcal{C})$ {cascade + Dijkstra}

return $\mathcal{C}'$

2.3 1/rank Normalization (Step 2)

Within each filtered layer, edges are sorted by weight descending and assigned $w_{ij} \leftarrow 1/\mathrm{rank}(w_{ij})$. This is scale-free: it preserves only the ordering, making heterogeneous layers (BC cosine similarity vs. DC fractional counts vs. Emb k-NN distance) directly comparable.

2.4 Consensus Sum (Step 3)

$$w_{ij}^{\text{cons}} = \left(\sum_{\ell=1}^{L} w_{ij}^{(\ell)}\right) \times n_{ij} \tag{1}$$

where $n_{ij} = |\{\ell : (i,j) \in E'_\ell\}|$ is the number of layers containing edge (i,j) after top-k filtering. Edges confirmed by multiple independent signals receive multiplicative amplification.

Rationale. In a 3-layer (BC+CC+DC) configuration:

- 73% of edges appear in only 1 layer (n_{ij}=1, no boost)
- 21% appear in 2 layers (n_{ij}=2, 2× boost)
- 6% appear in all 3 layers (n_{ij}=3, 3× boost)

The 6% of 3-layer edges form *cluster cores* in Leiden, while single-layer edges become relatively weak, preventing mega-cluster absorption.

2.5 Automatic Resolution Selection (Step 4)

Adding layers changes the effective edge density, shifting optimal γ non-linearly. We observe 24–32× shifts between 3-layer and 4-layer configurations—unpredictable from edge count ratios alone (4.2×). We therefore use binary search on log-scale γ to find the resolution where the largest cluster is $< \max_{\%}$ of total nodes (default 3%).

3 Experimental Setup

3.1 Data

Two OpenAlex fields with contrasting citation densities:

3.2 Comparison Methods

Normalization: raw, 1/rank, quantile.

Combination: sum ($\sum w$), max ($\max w$), vote (count), consensus ($\sum w \times n$).

Table 1: Dataset characteristics (top-30 filtered)

Layer	Field 15 (Chem. Eng.) Edges	Coverage	Field 12 (Arts & Hum.) Edges	Coverage
BC cosine	1.56M	86%	–	49%
CC cosine	524K	46%	–	18%
DC fractional	808K	100%	–	100%
Emb k-NN	2.86M	100%	19.3M	100%
DC avg degree	16		5	
Emb avg degree	50		51	

3.3 Evaluation

1. **Structural**: cluster count, max cluster %, size distribution balance.
2. **LLM blind review** (GPT-4o-mini, 55 cases): three criteria on disagreement nodes between sum and consensus:
   - *Belonging*: "Which group does this paper naturally belong to?"
   - *Group cohesion*: independent quality score per cluster (1–5).
   - *Outlier count*: papers that don't fit in the cluster.

4 Results

4.1 Normalization × Combination Grid

Table 2: Field 15, γ=10^{-6}, min_size=100, top-30, GCC. Cluster count and max cluster %.

Normalization	Combination	Clusters	Max %
raw	sum	189	12.2
raw	vote	149	15.5
1/rank	**sum**	**282**	**2.4**
1/rank	**consensus**	**298**	**2.5**
1/rank	vote	149	16.5
quantile	sum	156	15.8

1/rank with sum or consensus produces the most balanced distributions. Vote is invariant to normalization (binary). Raw sum creates dominant clusters.

4.2 Sum vs. Consensus: LLM Evaluation

Table 3: Triple evaluation on disagreement nodes (γ=10^{-4}, Field 15).

Criterion	Sum	Consensus	Better
Belonging (wins, n=55)	26	**29**	Consensus
Cohesion (avg, 1–5)	4.4	4.4	Tie
Outliers (avg per 8 neighbors)	1.3	1.3	Tie

Consensus assigns boundary nodes more accurately. Internal cluster quality is equivalent.

By cluster size bucket (30 cases):

- **Large (>500)**: Sum 7:3 — sum's mega-clusters win by volume.
- **Medium (200–500)**: Consensus 6:4 — consensus correctly separates.
- **Small (<200)**: Tie 5:5.

4.3 3-Layer vs. 4-Layer (+Embedding)

Table 4: Auto-gamma results (target max < 3%).

Config	Auto γ	Clusters	Max %	Avg size
Field 15, 3L	5.62×10^{-5}	251	3.0	459
Field 15, 4L	1.33×10^{-3}	221	2.8	521
Field 12, 3L	1.78×10^{-6}	1,157	2.9	651
Field 12, 4L	$\mathbf{5.62 \times 10^{-5}}$	**445**	**2.9**	**1,693**

Adding embedding edges consistently improves partitioning:

- Field 15: 12% fewer clusters, 13% larger avg size.
- Field 12: **62% fewer clusters, 160% larger avg size** — embedding fills citation gaps in humanities.

The γ ratio between 3L and 4L is 24× (Field 15) to 32× (Field 12), confirming that automatic search is essential.

4.4 Consensus Mechanism Analysis

The consensus/sum weight ratio is exactly $n_{ij} \in \{1, 2, 3, 4\}$:

Table 5: Edge distribution by layer count (Field 15, 3-layer).

n_{ij}	Fraction	Boost	Role
1	73%	1×	Boundary/noise
2	21%	2×	Supporting
3	6.5%	3×	**Cluster cores**

The 6.5% of 3-layer edges, amplified to 3× weight, dominate Leiden's objective function and form cluster nuclei. Single-layer edges become relatively weak, preventing the mega-cluster absorption observed with simple summation.

4.5 Temporal Edge Landscape

Each layer exhibits a distinct temporal pattern, visible in the year×year edge density heatmaps (Figure 1).

Table 6: Temporal concentration: fraction of edges within ±3 years of the diagonal.

Layer	Diagonal ±3yr	Asymmetry	Character
BC	78.0%	0.50	Same-era shared references
CC	73.7%	0.50	Slightly broader (classic papers)
Emb	62.3%	0.50	Text similarity $\propto$ era
DC	44.0%	0.50	Widest (past citations)

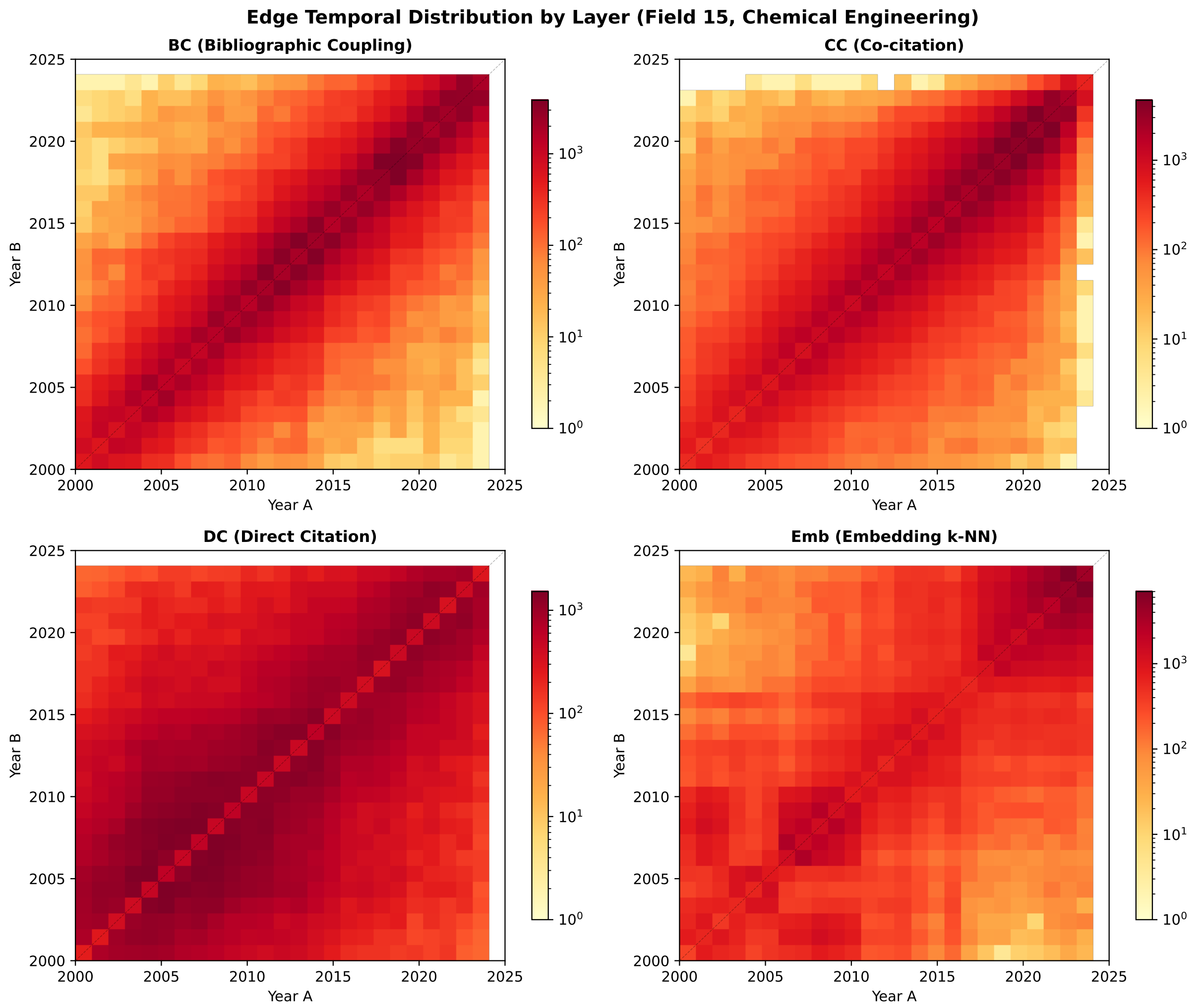

Figure 1: Year×year edge density (log scale) for four layers in Field 15. BC concentrates on the diagonal (same-era papers); DC spreads widely (citing past); CC lies between; Emb shows moderate diagonal concentration.

Stronger edges are temporally tighter: among BC edges, the top 1% by weight have 99.4% of connections within ±3 years (Figure 2). This confirms that consensus weighting preferentially amplifies temporally coherent, topically focused connections.

4.6 Consensus Level Analysis

Decomposing edges by consensus level reveals that higher consensus correlates with tighter temporal concentration (Figure 4).

The consensus weight distribution and temporal profile per level are shown in Figure 5.

Figure 6 decomposes the consensus by specific layer combinations.

Key patterns:

- **BC+CC is the strongest pair** (12,947 edges, 3.6%) — papers sharing references tend to also be co-cited.
- **Emb is the most independent layer** — lowest pairwise overlap with all citation layers (<7.3%), confirming it captures complementary information.
- **3-layer combinations involving BC+CC dominate** — BC+CC+Emb (1,479) and BC+CC+DC (1,353) are the most common triplets.

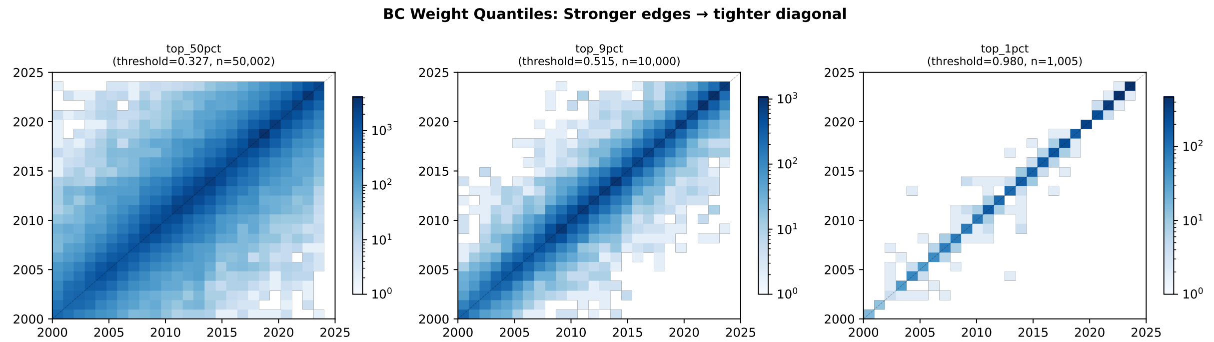

Figure 2: BC weight quantile maps: stronger edges concentrate increasingly on the diagonal. Top 1% edges connect almost exclusively same-era papers.

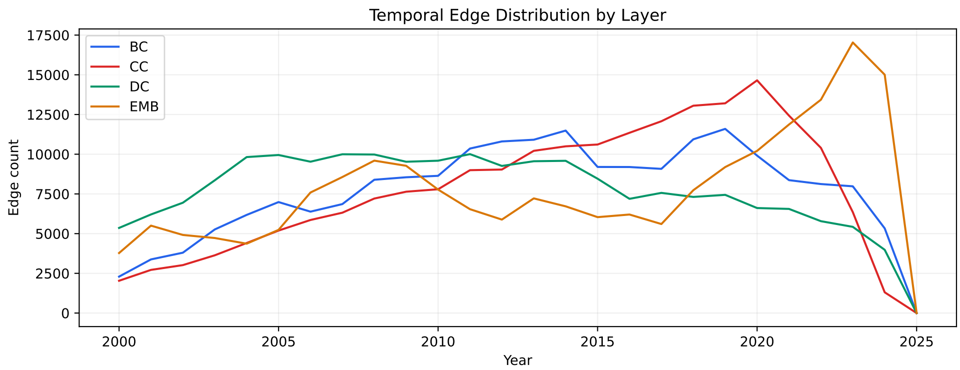

Figure 3: Temporal edge distribution (marginal) across layers. BC and CC peak in recent years; DC is flatter due to cumulative citations; Emb coverage is uniform.

- **Full consensus is rare but informative** — only 240 edges achieve 4-layer agreement, but these form the densest cluster cores.

5 Discussion

Why consensus works. CPM Leiden optimizes $H = \sum_c \left[e_c - \gamma\binom{n_c}{2}\right]$, where e_c is internal edge weight. Consensus weighting amplifies edges that multiple independent measures agree on, concentrating weight in true communities. This creates a natural hierarchy: multi-layer cores attract single-layer periphery.

Embedding layer integration. Embedding k-NN edges have universal coverage (avg degree ≈ 50) unlike citation layers (DC avg degree = 5 in humanities). Naïvely adding Emb to consensus increases n_{ij} globally, collapsing the differentiation. The solution is **not** to exclude Emb but to **automatically adjust** γ: auto-gamma search absorbs the density shift, letting Emb fill citation gaps while consensus preserves cluster structure.

Field dependence. Citation-poor fields (Arts & Humanities: 3-layer overlap 0.1%) benefit most from Emb addition (160% avg size improvement) because citation layers alone produce excessive fragmentation. Citation-rich fields (Chemical Engineering: 3-layer overlap 2.2%) see moderate improvement (13%).

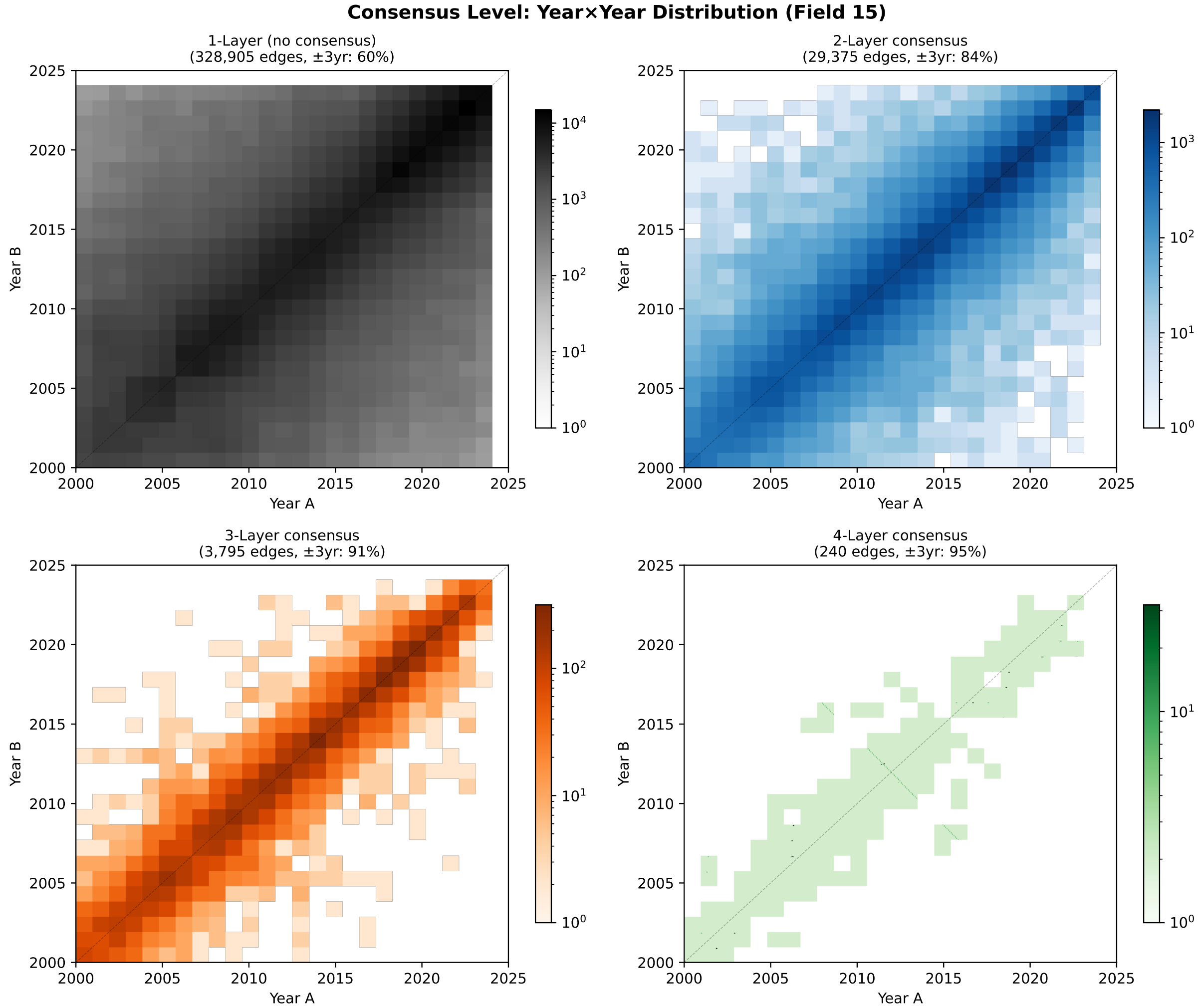

Figure 4: Year×year heatmaps by consensus level. 1-layer edges (91% of total) spread broadly; 3–4 layer edges concentrate sharply on the diagonal.

6 Conclusion

We recommend the following pipeline for multi-layer scientific paper clustering:

1. **Top-30** per-node filtering on each layer.
2. 1/**rank** normalization for scale-free comparison.
3. **Consensus weighting sum** ($\sum w \times n_{\text{layers}}$) for multi-layer agreement.
4. **GCC filtering** to remove isolated components.
5. **Auto-gamma** binary search targeting max cluster $< 3\%$.
6. **Leiden** + **postprocess** with cascading γ descent.

This pipeline adapts automatically to varying citation densities across fields and layer configurations, producing balanced cluster distributions without manual parameter tuning.

Table 7: Consensus level statistics (Field 15, 4 layers).

Consensus	Edges	Fraction	Diagonal ±3yr	Median w^{cons}
1-layer	328,905	90.8%	60%	≈ 0
2-layer	29,375	8.1%	84%	0.0001
3-layer	3,795	1.0%	91%	0.0005
4-layer	240	0.1%	95%	0.0012

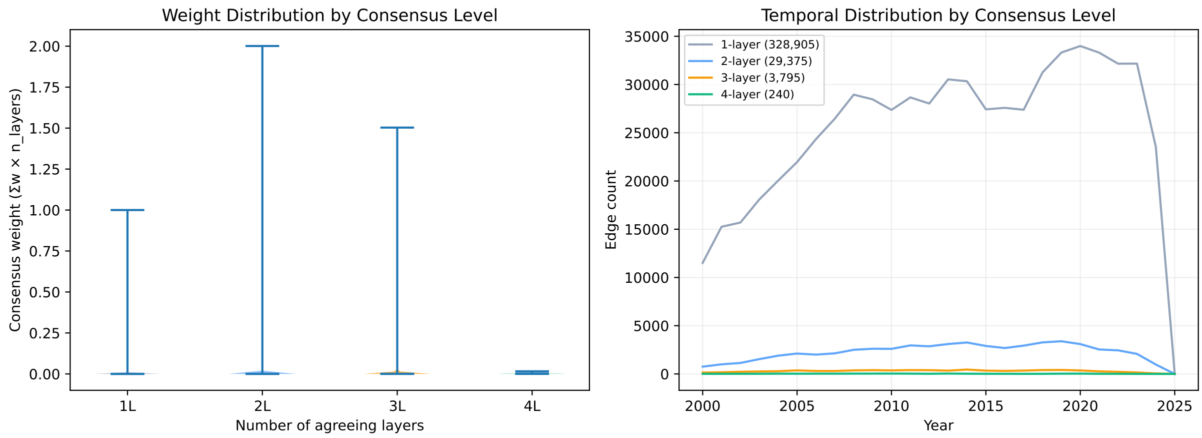

Figure 5: Left: consensus weight distribution (violin) by level—higher consensus produces heavier, more distinguishable weights. Right: temporal edge distribution per level—multi-layer edges peak in recent years where all data sources overlap.

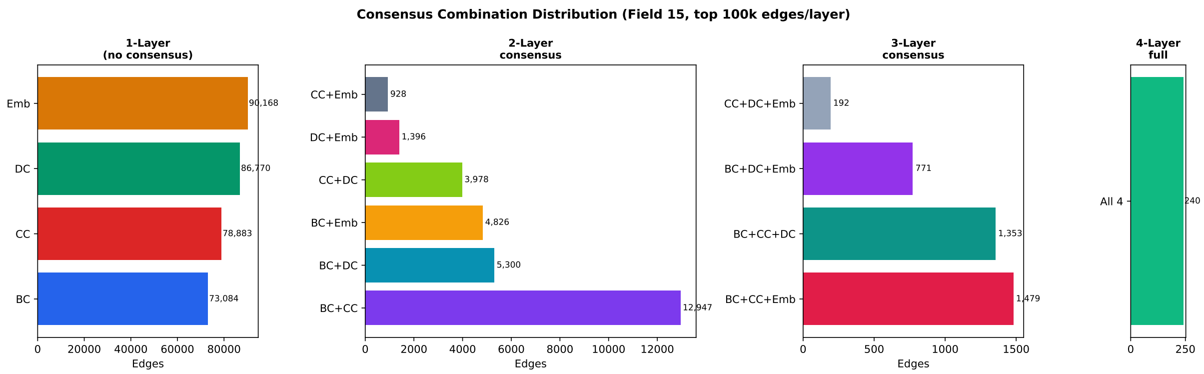

Figure 6: Edge count by consensus combination. Among 2-layer pairs, BC+CC dominates (3.6%); among 3-layer, BC+CC+Emb and BC+CC+DC are comparable. Only 240 edges (0.07%) achieve full 4-layer consensus.

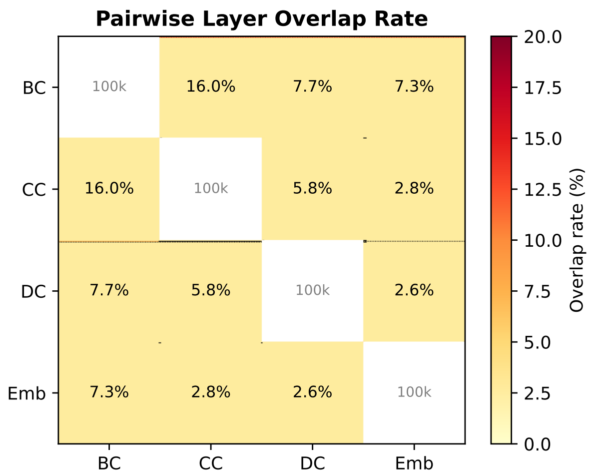

Figure 7: Pairwise overlap rate (% of smaller layer's edges). BC∩CC has the highest overlap (16%), while Emb overlaps least with citation layers (<7.3%).